

MSQF 536: MACHINE LEARNING TECHNIQUES

UNIT I ASSIGNMENT

Data Pre-processing

M.Sc. Quantitative Finance

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Section A: Conceptual Questions

[10 Marks]

1. Define Machine Learning in your own words. Explain why data pre-processing is an important stage in a machine learning workflow.
2. Distinguish between the following types of variables and give one suitable financial example for each:
 - (a) Continuous variable
 - (b) Discrete variable
 - (c) Nominal variable
 - (d) Ordinal variable
 - (e) Binary variable
3. Explain the difference between numerical and categorical variables. Why is this distinction important in data pre-processing?
4. Explain the following steps in a data pre-processing pipeline:
 - (a) Cleaning
 - (b) Transformation
 - (c) Normalization/Scaling
 - (d) Encoding
5. Explain the purpose of the following datasets:
 - (a) Training set
 - (b) Validation set
 - (c) Testing set

Section B: Numerical and Short Answer Questions [10 Marks]

1. Consider the following variables from a financial dataset:

{Stock Price, Number of Trades, Sector, Credit Rating, Loan Default}.

Classify each variable as continuous, discrete, nominal, ordinal, or binary.

2. The observed daily trading volumes of a stock are

100, 120, 140, 160, 180.

Calculate the mean value that would be used to replace a missing observation.

3. For

$$x = 80, \quad x_{\min} = 20, \quad x_{\max} = 100,$$

calculate the Min–Max scaled value using

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}.$$

4. A variable has mean $\mu = 50$ and standard deviation $\sigma = 10$. Calculate the Z-score when $x = 70$ using

$$z = \frac{x - \mu}{\sigma}.$$

5. A dataset contains 10,000 observations. If an 80–20 train–test split is used, determine the number of observations in the training and testing sets.
6. Explain why a random train–test split may be problematic for daily stock-price time-series data.
7. Consider the following confusion matrix:

	Predicted Positive	Predicted Negative
Actual Positive	40	5
Actual Negative	10	945

Calculate:

- (a) Accuracy
 - (b) Precision
 - (c) Recall
 - (d) F1-score
8. Explain why accuracy alone may be misleading when the target variable is highly imbalanced.

Section C: Analytical and Discussion Questions [12 Marks]

Answer any four questions.

1. Explain the difference between MCAR, MAR, and MNAR missing-data mechanisms. Give one financial example for each.
2. Compare the following missing-value imputation methods:
 - (a) Mean/Median imputation
 - (b) Mode imputation
 - (c) KNN imputation
 - (d) Forward fill
 - (e) Backward fill
 - (f) Interpolation
3. Explain the difference between normalization and standardization. Give the mathematical formula for each.
4. Explain one-hot encoding. Consider the categorical variable
$$\text{Sector} \in \{\text{Banking, IT, Energy, Pharma}\}.$$
Show how this variable can be represented using one-hot encoding.
5. Explain why KNN and SVM are sensitive to the scale of variables. Illustrate your explanation with a simple financial example.
6. Explain the difference between over-sampling and under-sampling. What is SMOTE, and how does it differ from simple duplication of minority observations?
7. Explain the concept of data leakage. Why is it incorrect to scale or impute the complete dataset before performing the train-test split?
8. Explain why fraud detection is a typical class-imbalance problem. Discuss why a model predicting “No Fraud” for every transaction may still achieve very high accuracy.

Section D: Python Exercises [10 Marks]

1. Identifying Variable Types

Create a Pandas DataFrame containing the following variables:

`stock, sector, credit_rating, close_price, num_trades.`

Using Python:

- (a) Display the DataFrame.
- (b) Display the data types.
- (c) Identify the categorical variables.
- (d) Identify the numerical variables.

2. Scaling and Encoding

Create a DataFrame containing a numerical variable `close_price` and a categorical variable `sector`.

Using Python:

- (a) Apply Min–Max scaling to `close_price`.
- (b) Apply Z-score standardization to `close_price`.
- (c) Apply one-hot encoding to `sector`.
- (d) Display the resulting DataFrame.

3. Missing-Value Imputation

Create a DataFrame containing at least two numerical variables with missing observations.

Using Python:

- (a) Identify the missing values.
- (b) Perform mean imputation.
- (c) Perform median imputation.
- (d) Perform KNN imputation.
- (e) Compare the results obtained from the three methods.

4. Train–Test Split

Using a suitable financial dataset, perform an 80–20 train–test split.

Report:

- (alph*) Number of observations in the original dataset.
- (alph*) Number of observations in the training set.
- (alph*) Number of observations in the testing set.
- (alph*) The Python code used.

5. Class Imbalance

Using a suitable binary classification dataset:

- (alph*) Display the class distribution before balancing.
- (alph*) Apply an appropriate sampling technique.
- (alph*) Display the class distribution after balancing.
- (alph*) Compare the class distributions before and after balancing.
- (alph*) Explain why the balancing procedure should not be applied to the test set.

Section E: Finance-Based Case Study [8 Marks]

Case Study: Loan Default Prediction

A financial institution has collected historical customer information to develop a machine learning model for predicting loan default. The dataset contains the following variables:

Variable	Description
Age	Customer age
Income	Annual income
Credit Score	Credit score
Employment Type	Employment category
Credit Rating	Rating category
Loan Amount	Amount of loan
Debt-to-Income Ratio	Ratio of debt to income
Previous Default	Previous default status
Loan Default	Target variable

Assume that the dataset contains missing values, categorical variables, numerical variables with different scales, and substantially fewer default cases than non-default cases.

Answer the following:

1. Identify the numerical and categorical variables.
2. Suggest appropriate encoding methods for the categorical variables.
3. Suggest suitable methods for handling missing values in the numerical and categorical variables.
4. Explain how the numerical variables should be scaled.
5. Propose a suitable train-validation-test strategy.
6. Explain how you would handle the class imbalance in the target variable.
7. Identify two possible sources of data leakage in this problem.
8. Prepare a step-by-step data pre-processing workflow for this dataset before applying a machine learning classification algorithm.